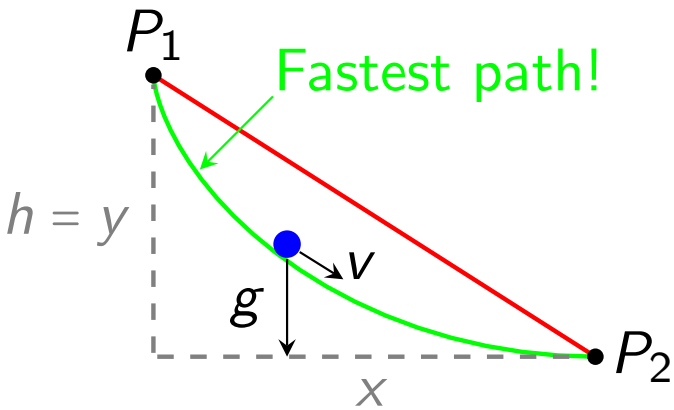
Lecture 07, Math 441

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Application: the Brachistochrone Problem

Find the curve along which a frictionless point of mass M slides under gravity between two points in the minimum amount of time. Assume the points are not vertically or horizontally aligned.



This is a frictionless system, so to solve this we use conservation of energy, where kinetic energy equals potential energy. $\frac{1}{2} M v^2 = M g h \rightarrow v = \sqrt{2gh}$

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Set y to be positive-downward so $h = y$ and $v = \sqrt{2gy}$.

Time to travel from P_1 to P_2 is

$$T = \int_{P_1}^{P_2} \frac{\mathrm{d}s}{v} \text{ where } \mathrm{d}s = \sqrt{(\mathrm{d}x)^2 + (\mathrm{d}y)^2} = \sqrt{1 + (y')^2}\,\mathrm{d}x.$$

Substituting and introducing $\mathcal{L}(x, y, y')$ for the integrand:

$$T = \int_{x_1}^{x_2} \frac{\sqrt{1 + (y')^2}}{\sqrt{2gy}}\,\mathrm{d}x = \int_{x_1}^{x_2} \mathcal{L}(x, y, y')\,\mathrm{d}x.$$

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If we change the path by a small amount δy, the time changes as

$$\delta T = \int_{x_1}^{x_2} \left(\frac{\partial \mathcal{L}}{\partial y}\, \delta y + \frac{\partial \mathcal{L}}{\partial y'}\, \delta y' \right) \mathrm{d}x$$

and using $\delta y' = \delta \left(\frac{\mathrm{d}y}{\mathrm{d}x} \right) = \frac{\mathrm{d}}{\mathrm{d}x}(\delta y)$:

$$\delta T = \int_{x_1}^{x_2} \left(\frac{\partial \mathcal{L}}{\partial y}\, \delta y + \frac{\partial \mathcal{L}}{\partial y'}\, \frac{\mathrm{d}}{\mathrm{d}x}(\delta y) \right) \mathrm{d}x.$$

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Use integration by parts on the second term:

$$\delta T = \int_{x_1}^{x_2} \frac{\partial \mathcal{L}}{\partial y}\,\delta y \mathrm{d}x + \left[\frac{\partial \mathcal{L}}{\partial y'}\delta y\right]_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial \mathcal{L}}{\partial y'}\right)\delta y \mathrm{d}x.$$

Since x_1 and x_2 correspond to the end points, $\delta y(x_1) = \delta y(x_2) = 0$

so the middle term is 0 and we have

$$\delta T = \int_{x_1}^{x_2} \left[\frac{\partial \mathcal{L}}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial \mathcal{L}}{\partial y'}\right)\right]\delta y\,\mathrm{d}x.$$

To find the minimum time, we set $\delta T = 0$.

To have this be true for any arbitrary δy, $\left[\frac{\partial \mathcal{L}}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial \mathcal{L}}{\partial y'}\right)\right] = 0$ for any $x_1 \leq x \leq x_2$.

This is called an Euler-Lagrange differential equation.

Let's write out

$$\frac{\mathrm{d}}{\mathrm{d}x}\mathcal{L}(x, y, y') = \frac{\partial \mathcal{L}}{\partial x} + \frac{\partial \mathcal{L}}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{\partial \mathcal{L}}{\partial y'}\frac{\mathrm{d}y'}{\mathrm{d}x} = \frac{\partial \mathcal{L}}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{\partial \mathcal{L}}{\partial y'}\frac{\mathrm{d}y'}{\mathrm{d}x}.$$

$\mathcal{L}$ does not explicitly depend on x, so $\frac{\partial \mathcal{L}}{\partial x} = 0$.

Then, multiply the Euler-Lagrange equation by y':

$$\frac{\partial \mathcal{L}}{\partial y}y' - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial \mathcal{L}}{\partial y'}\right)y' = 0.$$

Combining these two equations, we can write

$$\frac{\mathrm{d}}{\mathrm{d}x}\mathcal{L} - \frac{\partial \mathcal{L}}{\partial y'} y'' = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial \mathcal{L}}{\partial y'}\right) y',$$

which can then be rewritten as

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathcal{L} - \frac{\partial \mathcal{L}}{\partial y'}\, y'\right) = 0.$$

Integrate with respect to x:

$$\mathcal{L} - \frac{\partial \mathcal{L}}{\partial y'} y' = c \qquad \text{(Beltrami identity)}$$

Now, using

$$\mathcal{L} = \frac{\sqrt{1 + (y')^2}}{\sqrt{2gy}} \text{ and } \frac{\partial \mathcal{L}}{\mathrm{d}y'} = \frac{y'}{\sqrt{2gy}\sqrt{1 + (y')^2}},$$

we get (after some simplifications):

$$\frac{1}{\sqrt{2gy}\sqrt{1 + (y')^2}} = c \rightarrow \frac{1}{2gy\,[1 + (y')^2]} = c^2$$

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Rewrite

$$\frac{1}{2gy\left[1+(y')^2\right]} = c^2 \;\rightarrow\; y\left[1+(y')^2\right] = \frac{1}{2gc^2} = k^2$$

Solving for y':

$$\boxed{y' = \sqrt{\frac{k^2}{y} - 1}}$$

This is the equation we want to solve.

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Use the substitutions

$$y = k^2 \sin^2(t), \qquad \mathrm{d}y = 2k^2 \sin(t)\cos(t)\mathrm{d}t,$$

$$\frac{k^2}{y} - 1 = \frac{k^2}{k^2 \sin^2(t)} - 1 = \frac{1 - \sin^2(t)}{\sin^2(t)} = \frac{\cos^2(t)}{\sin^2(t)}.$$

The equation is now

$$2k^2 \sin(t) \cancel{\cos(t)} \mathrm{d}t = \frac{\cancel{\cos(t)}}{\sin(t)} \mathrm{d}x \;\rightarrow\; 2k^2 \sin^2(t) \mathrm{d}t = \mathrm{d}x.$$

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Using the double angle formula, $\sin^2(t) = \frac{1-\cos(2t)}{2}$:

$$2k^2\left(\frac{1-\cos(2t)}{2}\right)\mathrm{d}t = \mathrm{d}x \xrightarrow{\int} x = k^2\left(t - \frac{\sin(2t)}{2}\right) + A.$$

If we set P_1 at the origin, the constant of integration A can be set to zero (because then $x(0) = 0$). Similarly for y, we have

$$y = k^2\left(\frac{1}{2} - \frac{\cos(2t)}{2}\right).$$

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Setting $2t = \theta$, we get the equations

$$\boxed{\begin{aligned} x(\theta) &= \frac{k^2}{2}\left(\theta - \sin\left(\theta\right)\right) \\ y(\theta) &= \frac{k^2}{2}\left(1 - \cos\left(\theta\right)\right) \end{aligned}}$$

These parametric equations are the solution to the brachistochrone problem and describe a cycloid.

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